Annals of Epidemiology 104 (2025) 55–60

Contents lists available at ScienceDirect

Annals of Epidemiology

journal homepage: www.sciencedirect.com/journal/annals-of-epidemiology

Utilization of locally estimated scatterplot smoothing (LOESS) regression to estimate missing weights in a longitudinal cohort of breast cancer patients

Alexa Zimbalist *, Kelly H. Radimer, Isaac J. Ergas, Janise M. Roh, Charles P. Quesenberry, Marilyn L. Kwan, Lawrence H. Kushi

Division of Research, Kaiser Permanente Northern California, 4480 Hacienda Drive, Pleasanton, CA 94588, United States

ARTICLE INFO

Keywords:
Missing data
Locally estimated scatterplot smoothing
Longitudinal data
Weight data
Breast cancer

ABSTRACT

Introduction: Traditional methods to handle missing data rely on making assumptions about missing data patterns. Locally estimated scatterplot smoothing (LOESS) regression models were explored as a data-driven option to minimize missing weight data in a longitudinal cohort of breast cancer patients.

Methods: Outpatient weights from 2 years prior to breast cancer diagnosis to 10 years post were extracted from electronic health records for 10,778 women with invasive breast cancer diagnosed from 2005-2013 at Kaiser Permanente. LOESS regression models estimated weights at baseline (breast cancer diagnosis) and 6 follow-up time points (6, 12, 24, 48, 72, and 96 months post-baseline). The weights identified by the LOESS models were compared with those identified by the closest-available method, in which the weight measurement closest to each timepoint within a specified time window was selected.

Results: Compared with the closest-available method, LOESS models identified fewer weights at baseline and 6 months post, but significantly more weights at later follow-up periods. At all timepoints, more than 80% of the weights identified by both approaches differed by 2.50 kilograms or less.

Conclusions: LOESS regression makes effective use of available longitudinal data and may be a beneficial tool to minimize missing longitudinal data in future EHR-based studies.

Introduction

Breast cancer is the most common cancer in women in the United States and the relationship between obesity and breast cancer risk and outcomes has been widely studied [1]. Weight data play an important role in breast cancer research, particularly in longitudinal studies that aim to understand how risk factors impact disease onset and prognosis. Yet, similar to other kinds of questionnaire and electronic health record (EHR) data, weight data can often be missing when patients decline to self-report their weight, miss medical appointments, or are lost to follow-up.

Addressing missing data in epidemiologic analyses is important to minimize loss of precision and avoid biased study results [2]. While several approaches to handle missing data exist, such as complete case analysis, missing indicators, and multiple imputation, they all have their advantages and disadvantages in producing valid analyses [2,3]. The approach chosen to handle missing data in analyses is typically dependent on whether the data are assumed to be missing completely at random (MCAR), missing at random (MAR), or missing not at random (MNAR) [3]. Complete case analysis, in which only individuals with non-missing data are analyzed, relies on the assumption that the data are MCAR, and may lead to a loss of power and non-representative results [4]. Creating a missing indicator for missing data has been criticized for resulting in biased estimates, and misspecification may occur when using multiple imputation models, in which missing data values are replaced by plausible values based on other characteristics of study participants [5,6].

Rather than making assumptions about missing data patterns and relying on conventional methods to handle missing data, we explored the application and effectiveness of locally estimated scatterplot smoothing (LOESS) regression models to minimize missing weight data in a longitudinal cohort of women with breast cancer. LOESS regression is a non-parametric smoothing tool that aims to fit a smooth curve through a scatterplot of data points [7]. We aimed to use available EHR data to fit LOESS regression models to impute each individual's missing weight data and illustrate a potential technique to build more robust

* Corresponding author.
E-mail address: alexa.x.zimbalist@kp.org (A. Zimbalist).

https://doi.org/10.1016/j.annepidem.2025.02.013
Received 17 October 2024; Received in revised form 20 February 2025; Accepted 24 February 2025
Available online 4 March 2025
1047-2797/© 2025 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

longitudinal datasets in the future.

Methods

Study population

This study population was drawn from Kaiser Permanente Northern California (KPNC) health plan members who were eligible for the Pathways Study, a prospective cohort study that examines how lifestyle, molecular, clinical, and neighborhood factors impact breast cancer outcomes [8]. The Pathways Study began participant recruitment in January 2006 and identified a total of 11,173 women 21 years and older who were diagnosed with invasive breast cancer (all stages) between November 2005 and May 2013, of whom 4,504 women were enrolled. Participants were interviewed in-person at baseline, which was approximately two months after cancer diagnosis, and at 6, 12, 24, 48, 72 and 96 months after baseline by mail, phone, online, or in-person. Of the women eligible for the Pathways Study (n=11,173), women were included in this study if they had at least 10 outpatient non-outlier weight measurements from the KPNC EHR between 2 years before breast cancer diagnosis and 10 years after diagnosis (n=10,778). All methods in this study were conducted in accordance with relevant guidelines and regulations approved by the KPNC Institutional Review Board.

Weight measurements

We extracted 549,225 outpatient weight measurements (kilograms) between 2 years prior to breast cancer diagnosis and 10 years post-diagnosis from the EHR. Of these weights, 707 outlier weights were identified for exclusion using the growthcleanr R package, which compares the standard deviation score for each weight to a weighted moving average to flag values with large deviations from the expected value [9]. In order to minimize sparse data, only women with 10 or more weight measurements were retained after excluding outliers, resulting in 10, 778 women with a total of 546,576 weight measurements (Fig. 1). Women excluded due to insufficient weight data were similar to those included with respect to age, race, body mass index, and breast cancer characteristics.

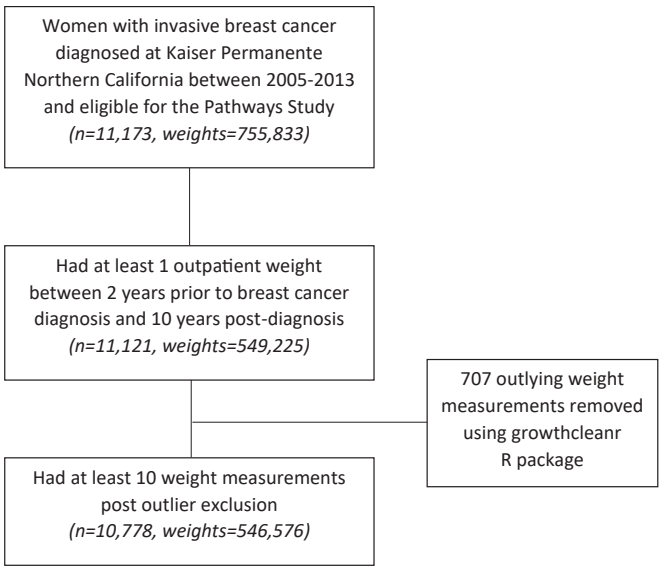

Fig. 1. Cohort description.

Identification of baseline and follow-up weight measurements

Field data collection for the Pathways Study occurred at baseline and 6, 12, 24, 48, 72 and 96 months post-baseline and thus, subsequent analyses have focused on data from these timepoints. Because this study population includes Pathways Study participants, we utilized breast cancer diagnosis date as the baseline timepoint and the same 6 follow-up timepoints as periods of interest. The 546,576 EHR weight measurements were used to derive a weight estimate at baseline and each of the follow-up timepoints using two methods described below: (1) Closest-available weight selection method and (2) LOESS method.

Closest-available weight selection method

Traditionally, Pathways Study researchers have prioritized EHR anthropometric data over weight data from interviews and questionnaires in order to minimize response bias [10]. This study utilized EHR data only and employed the closest-available weight selection method commonly used in prior Pathways Study analyses and publications. The method identifies the breast cancer diagnosis (baseline) weight as the outpatient weight measurement closest to breast cancer diagnosis date within the window from 180 days prior to diagnosis until 90 days post-diagnosis or date of definitive surgery, whichever occurred earliest. Weights for each of the follow-up timepoints are selected as the weight measurement closest to the timepoint and within +/-90 days.

LOESS method

LOESS regression models were fit for each individual using their available weight measurements. LOESS regression is a non-parametric technique that does not make assumptions about the structure of the data and uses local polynomial models on a moving collection of data points [7]. The technique relies on a span to be identified, which is the proportion of points used for each of the local regressions. The range for possible spans is 0 to 1, with smaller spans resulting in closer fitting to the data and less smooth regression curves, while larger spans are typically used when fewer data points are available to ensure a smoother curve [7,11]. All LOESS models were created with first degree polynomials using the loess.as function from the R package fANCOVA, which automatically selects a span according to the Akaike Information Criterion (AIC) (RStudio version 4.2.1). The LOESS models were subsequently used to estimate weights for the dates of interest, namely diagnosis date (baseline) and the 6 follow-up dates. Only LOESS estimated weights with defined standard errors were included.

The weights identified by the closest-available weight selection method and estimated by the LOESS method were compared to understand the magnitude of differences at the different time points of interest. Additionally, the methods were assessed by the frequency of identified weights at each timepoint. Because the LOESS method makes greater use of available weight data, we hypothesized that if the weights identified by the two methods are similar, the estimated weights from the LOESS method may serve as a plausible option for minimizing missing weights over time.

Results

Patient characteristics are shown in Table 1. The mean (standard deviation (SD)) age at diagnosis was 60 (12.6) years. Most cohort members were non-Hispanic white women (60%), and most were diagnosed with stage 1 (52%) and ER-positive breast cancers (83%). A total of 546,576 EHR-based outpatient weight measurements were identified between 2 years prior to breast cancer diagnosis and 10 years post-diagnosis. The number of measurements per person ranged from 10 to 340, with a mean (SD) of 66 (36). The timing of the weight measurements relative to breast cancer diagnosis is shown in Fig. 2. Nearly half (47%) of the weight measurements occurred between breast cancer diagnosis and 3 years post-diagnosis.

In total, the LOESS models estimated 64,417 weight measurements,

Table 1
Study population characteristics at diagnosis (n=10,778)[1].

Age at diagnosis (years)	60.04 (12.55)
Race/ethnicity	
White	6498 (60%)
Black	994 (9.2%)
Asian	1919 (18%)
Hispanic	1182 (11%)
American Indian/Alaska Native	121 (1.1%)
Pacific Islander	62 (0.6%)
Missing	2 (<0.1%)
Body Mass Index (kg/m^2) at diagnosis	
< 18.5	135 (1.3%)
18.5-< 25	3313 (31%)
25-< 30	3390 (31%)
30-< 35	2004 (19%)
> =35	1653 (15%)
Missing	283 (2.6%)
Overall mortality	2941 (27%)
Breast cancer mortality	978 (9.1%)
ER positive	8896 (83%)
PR positive	6828 (63%)
HER2 positive	1454 (13%)
AJCC stage at diagnosis	
I	5639 (52%)
II	3787 (35%)
III	1046 (9.7%)
IV	264 (2.4%)
Missing	42 (0.4%)
Surgery	10,323 (96%)
Radiation	4448 (41%)
Chemotherapy	4534 (42%)
Hormonal therapy	7096 (66%)

[1]Mean (SD); n (%)

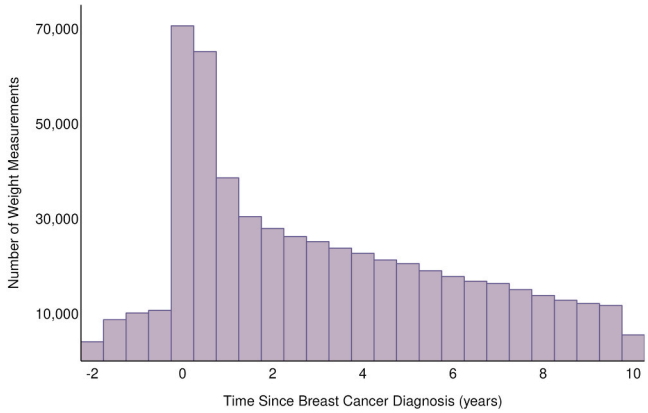

Fig. 2. Distribution of initial weight measurements.

of which 132 had undefined standard errors and were subsequently removed, resulting in 64,285 weights. Across all LOESS weights, the mean (SD) standard error was 0.51 (0.31). Table 2 presents summary statistics of the standard errors associated with the LOESS estimated weights at each time point. The minimum and maximum spans used for the LOESS regression models was 0.05 and 0.94, with a mean (SD) of 0.47 (0.24). Fig. 3 shows the distribution of spans selected for the LOESS models. The number of women with weight measurements identified by the closest-available weight selection method, the LOESS method, and both methods at baseline and each of the 6 follow-up timepoints is presented in Table 3. At baseline, 92% of women had an available weight as identified by the closest-available weight selection method, whereas 84% had an available baseline weight as estimated by the LOESS method. The LOESS method was able to estimate more weight measurements at later follow-up times, particularly 48, 72, and 96 months post-baseline. For example, at 96 months post-baseline, the closest-available weight selection method identified weights for 51% of the cohort, in comparison to 69% for the LOESS method. At 6 months post-baseline, 91% of women had a weight measurement identified by both methods, but the frequency of weight measurements decreased over time, with only 49% of people having a weight measurement identified by both methods at 96 months post-baseline.

The weight measurements from the closest-available weight selection method and LOESS method were highly correlated, with a Spearman's correlation of 0.99 across all timepoints. Table 4 presents the differences in weights for women who had measurements from both methods at each timepoint. The mean (SD) difference, in kilograms, between the two measurements varied from 1.0 (1.0) at 6 months post-baseline to 1.3 (1.4) at 96 months post-baseline. The maximum difference between the two measurements was 38.7 kilograms at 96 months post-baseline. Additionally, as shown in Table 5, at every timepoint more than 80% of the weights identified by the two methods differed by 2.50 kilograms or less.

Discussion

This study of 10,778 women with invasive breast cancer explored the possibility of using LOESS regression as a tool to minimize missing data. We found that such local regression models allow the leveraging of rich EHR data to estimate missing weights at specified longitudinal timepoints of interest.

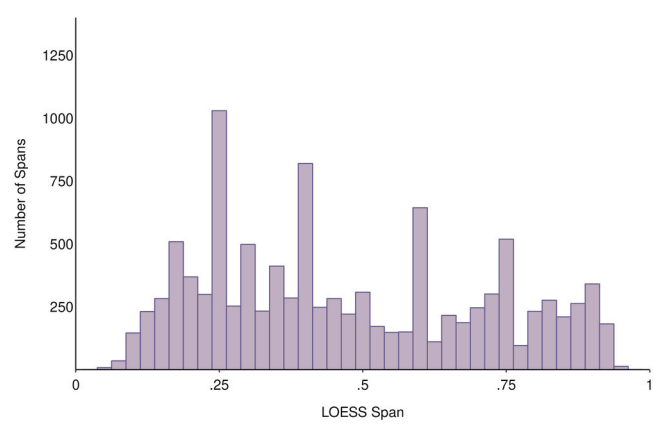

Fig. 3. LOESS regression spans.

Table 2
Standard errors associated with LOESS estimated weights.

	Baseline (breast cancer diagnosis) (n = 9078)	6 months post-baseline (n = 9990)	12 months post-baseline (n = 10,189)	24 months post-baseline (n = 10,040)	48 months post-baseline (n = 9205)	72 months post-baseline (n = 8392)	96 months post-baseline (n = 7391)
Minimum	0.11	0.11	0.10	0.10	0.11	0.12	0.13
Median	0.46	0.40	0.44	0.44	0.43	0.43	0.48
Mean	0.52	0.47	0.51	0.52	0.51	0.51	0.56
Standard Deviation	0.27	0.26	0.30	0.33	0.35	0.34	0.33
Maximum	4.47	4.48	5.45	7.16	7.97	8.69	6.41

Table 3
Number (%) of identified weight measurements (n=10,778).

	Baseline (breast cancer diagnosis)	6 months post-baseline	12 months post-baseline	24 months post-baseline	48 months post-baseline	72 months post-baseline	96 months post-baseline
Closest-available weight selection method	9907 (92%)	10,082 (94%)	9942 (92%)	9323 (87%)	8094 (75%)	6687 (62%)	5446 (51%)
LOESS method	9078 (84%)	9990 (93%)	10,189 (95%)	10,040 (93%)	9205 (85%)	8392 (78%)	7391 (69%)
Both methods	9075 (84%)	9803 (91%)	9714 (90%)	9173 (85%)	7959 (74%)	6573 (61%)	5293 (49%)

Table 4
Differences between weight measurements (kilograms) identified by closest-available weight selection method and LOESS method.

	Baseline (breast cancer diagnosis) (n = 9075)	6 months post-baseline (n = 9803)	12 months post-baseline (n = 9714)	24 months post-baseline (n = 9173)	48 months post-baseline (n = 7959)	72 months post-baseline (n = 6573)	96 months post-baseline (n = 5293)
Minimum	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Median	0.7	0.7	0.8	0.8	0.9	0.9	0.9
Mean	1.0	1.0	1.0	1.1	1.2	1.2	1.3
Standard Deviation	1.0	1.0	1.0	1.1	1.2	1.3	1.4
Maximum	19.4	15.9	22.8	13.0	15.5	24.6	38.7

Table 5
Categorical differences between weight measurements (kilograms) identified by closest-available weight selection method and LOESS method.

	Baseline (breast cancer diagnosis) (n = 9075)	6 months post-baseline (n = 9803)	12 months post-baseline (n = 9714)	24 months post-baseline (n = 9173)	48 months post-baseline (n = 7959)	72 months post-baseline (n = 6573)	96 months post-baseline (n = 5293)
< 0.50 kg	3397 (37%)	3401 (35%)	3403 (35%)	3042 (33%)	2554 (32%)	2024 (31%)	1585 (30%)
0.50 – 2.50 kg	5091 (56%)	5746 (59%)	5622 (58%)	5338 (58%)	4601 (58%)	3756 (57%)	3060 (58%)
2.51 – 4.50 kg	495 (5%)	574 (6%)	592 (6%)	661 (7%)	652 (8%)	632 (10%)	501 (9%)
> 4.50 kg	92 (1%)	82 (1%)	97 (1%)	132 (1%)	152 (2%)	161 (2%)	147 (3%)

Missing data occur in nearly every epidemiological study, yet how such data are handled varies greatly [12]. Understanding the missing data mechanism is an important first step prior to choosing the appropriate technique to minimize missingness. Missing data mechanisms include missing completely at random (MCAR), missing at random (MAR), and missing not at random (MNAR) [3]. Missing completely at random means the probability of having a variable with missing data does not depend on any observed variables. While this is the strongest assumption a researcher can make regarding the missing data mechanism, it is typically the most unrealistic [12]. Data are missing at random when the probability of missingness is dependent on other observed variables, while data that are missing not at random means the missingness pattern depends on unobserved variables [12]. Given the observed data, it is not possible to identify whether the data mechanism is missing at random or missing not at random [13]. Thus, choosing the missing data mechanism relies on making assumptions about the data (that ideally should be assessed via sensitivity analyses) [13].

Once an assumption has been made about the missingness pattern, a variety of approaches may be used to handle the missing data. Two common approaches in epidemiologic studies are complete case analysis and multiple imputation. Complete case analysis, the most widely used method to deal with missing data, relies on the inclusion of individuals with non-missing data only and generally yields unbiased results if the data are MCAR, which is often not the case in epidemiologic studies [4–6,12]. Moreover, in studies that rely on EHR data, it is often not the case that the data are MCAR, as most health encounters occur because of a clinical reason, such as a complaint or receipt of treatment. Multiple imputation, in which several different imputed data sets are created and the results are combined, is a more sophisticated approach to handling missing data, yet poses significant room for error when specifying the models [13].

These traditional methods not only require making assumptions about the structure of the missing data, but are also only applicable when there is a specified research question, with identified exposure, outcome and covariate variables. For example, in complete case analysis, participants are excluded based on missingness of covariate variables, and in multiple imputation models, it is best practice to include the outcome variable in the imputation models [4,14]. The Pathways study data is curated by KPNC and used in subsequent analyses performed by academic partners across the country. Traditionally, the Pathways study team has relied on the closest-available weight selection method to identify longitudinal weights. We were interested in an alternative method that could more effectively minimize missing weight data without having to identify a specific research question.

In breast cancer research, it is common to be interested in available measurements at or near certain clinically relevant timepoints, such as before, during and after cancer treatment. The closest-available weight selection method relies on a "window" of 90 days to identify weight measurements at a given timepoint. For example, to identify the weight of a patient at 6 months-post cancer diagnosis, the weight measurement closest to this timepoint would be selected as long as it was within +/- 90 days of the 6 months timepoint. We hypothesized that for women without an available weight measurement during the established "window," we could potentially use an average of the weight measurements in the more distant vicinity of the timepoint to estimate the weight at the desired timepoint. This hypothesis led us to examine the application of locally estimated scatterplot smoothing (LOESS) regression models, which fit curves and surfaces to data in a moving fashion without making assumptions about the structure of the data [7,15]. Such non-parametric local regressions provided an opportunity to utilize all existing longitudinal weight data, not just data within a pre-defined window, to estimate missing weights at timepoints of interest.

As hypothesized, compared with the closest-available method employed in prior Pathways studies, LOESS models could incorporate

information from all available weight measurements and thus resulted in more estimated weights in this study at every time point except at baseline and 6 months post-baseline. Moreover, when comparing the weight measurements identified by the closest-available method and estimated by the LOESS method, more than 80% of the measurements differed by 2.50 kilograms or less, suggesting that the LOESS method can serve as a reasonable way to identify an increased number of longitudinal weight measurements.

This study highlights that the LOESS method can serve as a tool to minimize missingness in longitudinal studies without having to make assumptions about the structure of the data. However, similar to other methods that handle missingness, LOESS models have limitations. Because LOESS employs local regressions on a moving collection of datapoints (called a neighborhood), it works best with dense data sets [7]. A data set with few data points, high dispersion of such points, or many outliers is not a good candidate for LOESS and may result in estimations with undefined standard errors. In such cases, alternative methods to handle missing data should be considered.

In this study, for the LOESS models, we used the R package fANCOVA which automatically selects a span, the size of the neighborhood over which averaging takes place [11]. Choosing a span is an important step in the LOESS modeling process, and incorrectly specifying a span for a dataset may result in predictions that are distant from the overall trend of the data points. For example, if a small span is applied to a dataset with few data points, the resulting LOESS curve may appear jagged and more closely follow the individual data points rather than the overall trend [7,11]. Additionally, although we know the span used for each individual in this study, we do not know the range of time points used for the local regressions. For example, given an individual with more dispersed, rather than dense weight measurements, the weights used to construct the local regression at 6 months post-baseline could be quite distant from that time point. Yet, the strong correlation between the longitudinal weight measurements identified by the closest-available and LOESS methods provide assurance that the weights estimated by the LOESS models are reasonable.

LOESS is a non-parametric technique that makes no assumptions about the structure of the data and thus, it cannot be used to estimate measurements outside the bounds of the available data. For this reason, we see fewer weight measurements predicted by LOESS at baseline (breast cancer diagnosis), in comparison to the closest-available method. This is because for 800 women, their first available weight measurement in the EHR occurred after breast cancer diagnosis. If the measurement occurred within 90 days post-cancer diagnosis that weight was identified as the baseline weight by the closest-available method. However, the LOESS model was not able to estimate a baseline (at-diagnosis) weight, because that timepoint was beyond the bounds of the first available data point.

As with the use of any missing data method, it is important to consider the underlying reasons for missingness. In this analysis, we hypothesized that the weight data were missing at random. We assumed the missingness to be systematically related to cancer stage and treatment, as those with advanced cancer and more frequent treatments likely visited the clinic setting more often and had fewer missing weight measurements. This paper presents a method to minimize missing data prior to answering a research question, but future studies should explore the association between weight and breast cancer outcomes. Sensitivity analyses should be conducted to assess missing data mechanism assumptions, as well as how utilizing LOESS estimated data may change observed relationships with breast cancer outcomes.

This study was limited to the application of LOESS to weight data, which generally tend to not change drastically over short periods of time. With such data, we expect that the LOESS local regressions are typically applied within a fairly narrow range of values. Alternatively, if LOESS were applied to other clinical metrics that have large fluctuations within short time periods, such as blood pressure or continuous glucose measurements, we would expect the local regressions to be applied within a much wider range of values and the LOESS estimated measurements to be averages that may be much further from a given observed data point. We postulate a likely limitation of LOESS is that it may not be an effective tool for measurements with large amounts of variation over short periods of time. In addition to its use with weight data, LOESS may serve as a helpful tool for other clinical measurements with typically less variation, such as hemoglobin A1C and prostate-specific antigen.

Often when data are missing, nearby longitudinal data points are not available to leverage and utilize in LOESS models. However, when feasible, the LOESS method employed in this study represents a potential tool that minimizes missingness in longitudinal cohorts without needing to specify a research question and make assumptions about the nature of the missing data. With the increasing interest in the use of clinical databases and real-world data for studies of disease risk, prognosis, and treatment effectiveness, this approach may potentially be applied to other clinical measurements that are documented repeatedly in clinical databases.

Conclusion

This study suggests that LOESS regression may be a beneficial tool to minimize missing longitudinal weight data without needing to make assumptions about the structure of the missing data. Such local regression models allow the leveraging of rich EHR data to understand individual weight trajectories and enable estimation of weights at specified longitudinal timepoints of interest. Future studies should address whether LOESS regression may be applied to minimize other types of missing data, such as laboratory measurements, in longitudinal studies that rely on data from electronic health records.

Funding

This work was supported by the National Cancer Institute [grant numbers U01CA195565, R01CA105274].

CRediT authorship contribution statement

Zimbalist Alexa: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Data curation. **Radimer Kelly H.:** Writing – review & editing, Formal analysis, Data curation, Conceptualization. **Ergas Isaac J.:** Writing – review & editing, Methodology, Conceptualization. **Roh Janise M.:** Writing – review & editing, Project administration. **Quesenberry Charles P.:** Writing – review & editing, Methodology. **Kwan Marilyn L.:** Writing – review & editing, Methodology, Conceptualization. **Kushi Lawrence H.:** Writing – review & editing, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Lawrence Kushi reports financial support was provided by National Cancer Institute. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] Playdon MC, Bracken MB, Sanft TB, Ligibel JA, Harrigan M, Irwin ML. Weight gain after breast cancer diagnosis and all-cause mortality: systematic review and meta-analysis. J Natl Cancer Inst 2015;107(12):djv275. https://doi.org/10.1093/jnci/djv275.

[2] Sterne JA, et al. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ 2009;338. https://doi.org/10.1136/bmj.b2393.

[3] Heymans MW, Twisk JWR. Handling missing data in clinical research. J Clin Epidemiol 2022;151:185–8. https://doi.org/10.1016/j.jclinepi.2022.08.016.

[4] Groenwold R, White I, Donders R, Carpenter J, Altman D, Moons K. Missing covariate data in clinical research: when and when not to use the missing-indicator method for analysis. CMAJ 2012;184(11):1265–9. https://doi.org/10.1503/cmaj.110977.
[5] Donders AR, van der Heijden GJ, Stijnen T, Moons KG. Review: a gentle introduction to imputation of missing values. J Clin Epidemiol 2006;59(10):1087–91. https://doi.org/10.1016/j.jclinepi.2006.01.014.
[6] Pedersen AB, et al. Missing data and multiple imputation in clinical epidemiological research. Clin Epidemiol 2017;9:157–66. https://doi.org/10.2147/CLEP.S129785.
[7] Turicchi J, et al. Data ImputatIon and Body Weight Variability Calculation Using Linear and Nonlinear Methods in Data Collected from Digital Smart Scales: Simulation and Validation Study. JMIR Mhealth Uhealth 2020;8(9):e17977. https://doi.org/10.2196/17977.
[8] Kwan ML, et al. The Pathways Study: a prospective study of breast cancer survivorship within Kaiser Permanente Northern California. Cancer Causes Control 2008;19(10):1065–76. https://doi.org/10.1007/s10552-008-9170-5.
[9] Daymont C, Ross ME, Russell Localio A, Fiks AG, Wasserman RC, Grundmeier RW. Automated identification of implausible values in growth data from pediatric electronic health records. J Am Med Inf Assoc 2017;24(6):1080–7. https://doi.org/10.1093/jamia/ocx037.
[10] Rosenman R, Tennekoon V, Hill LG. Measuring bias in self-reported data. Int J Behav Health Res 2011;2(4):320–32. https://doi.org/10.1504/IJBHR.2011.043414.
[11] Friedman JH. A variable span smoother. Submitt J Am Stat Assoc 1984.
[12] Perkins NJ, et al. Principled approaches to missing data in epidemiologic studies. Am J Epidemiol 2018;187(3):568–75. https://doi.org/10.1093/aje/kwx348.
[13] Sterne J, et al. Multiple imputation for missing data in epidemiological and clinical research: potential and pitfalls. BMJ 2009;338. https://doi.org/10.1136/bmj.b2393.
[14] Austin PC, White IR, Lee DS, van Buuren S. Missing data in clinical research: a tutorial on multiple imputation. Can J Cardiol 2021;37(9):1322–31. https://doi.org/10.1016/j.cjca.2020.11.010.
[15] Cleveland W, Devlin S, Grosse E. Regression by local fitting: methods, properties, and computational algorithms. J Econ 1988;37(1):87–114. https://doi.org/10.1016/0304-4076(88)90077-2.